Text and Unstructured Data

ECON 4370 / BANA 4373 Applied Data Tools for Economics & Business

Agenda (and where we are in the weekly schedule)

This lecture corresponds to: Text and Unstructured Data.

Today we will:

- Define unstructured data and explain why economists care about it
- Walk through the complete **text-to-numbers pipeline** step by step
- Build **word frequency** and **TF-IDF** measures from FOMC statements
- Apply **VADER sentiment analysis** to produce a monetary policy arc
- Demonstrate where general-purpose NLP tools **fail** on financial text

Big picture: Every regression you run needs numbers in the X matrix. This lecture shows how to get numbers from words.

The Data Landscape

Structured Data

- Rows and columns
- Known types and ranges
- FRED, BLS, Census ACS
- Ready for regression

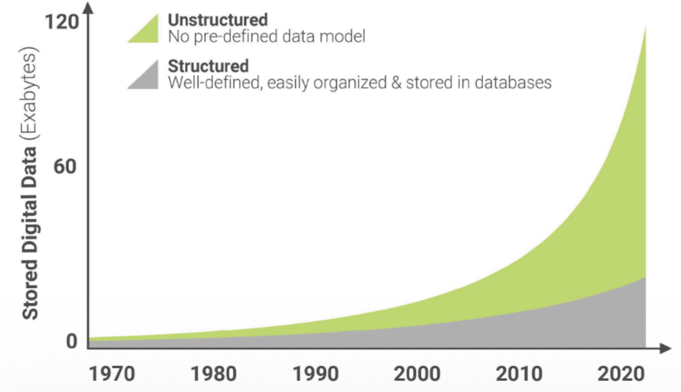
Unstructured Data

- No rows, no columns
- Human language
- Fed statements, 10-Ks, news
- Must be **converted** first

$\sim$80% of enterprise data is unstructured.

Most economists have only been trained to analyze the other 20%.

Structured vs Unstructured Data



Other Types of Data

Happenstance Data

- Data not collected for research purposes
- Exists because of everyday activity
- Repurposed for analysis

Definition

Data that was not intentionally designed for research, but can still be used to generate insights.

	Administrative	Happenstance
Structured	Traditional Economic Data	Credit Card Transactions Amazon product ratings
Unstructured	10-K Filings FOMC press conferences	Tweets Online Job Postings

Examples

- Online reviews (IMDb, Amazon)
- Company transaction data
- Social media posts

Text Data in This Course

- In this course, we use **text data** to illustrate unstructured data
- Text is the **most common form** of unstructured data in economics and finance
- Many core challenges generalize beyond text

Key Assumption

We abstract from how text is generated and assume it is organized into:

- Coherent documents
- Associated metadata (time, source, firm, etc.)

Measurement Problems in Text

1. **Concept presence** Is something in the document? (e.g., sentiment)
2. **Concept relationships** How do ideas relate? (e.g., sentiment toward firms)
3. **Distance between documents** How similar are documents?
4. **Linking to metadata** How do documents map to outcomes? (e.g., recession vs expansion)

What is Text?

Definition

Text is simply a **string of characters**

- Latin letters (a, A, p)
- Decorated letters (ö)
- Non-Latin characters (Chinese, Arabic)
- Punctuation (!, ?)
- Spaces, tabs, newlines
- Numbers
- Special characters (@, #, $)

Key Question

How do we convert text into a **quantitative representation**?

What is NLP?

Natural Language Processing (NLP)

A set of methods used to analyze, understand, and extract information from human language.

- Text data is unstructured and not directly usable in economic analysis
- NLP converts text into quantitative representations

Examples in Economics

- Sentiment analysis (e.g., IMDb reviews, financial news)
- Central bank communication (FOMC statements)
- Earnings calls and firm disclosures

From Text to Data

- Goal: convert text into **units of meaning**
- Typically: words or tokens
- First step: **pre-processing**

Important

We will cover pre-processing (cleaning, tokenization, etc.) in the practical session.

Notation

- Corpus: D documents indexed by d
- Document d: sequence of terms

$$w_d = (w_{d,1}, \dots, w_{d,N_d})$$

- Total terms: $N = \sum_d N_d$
- Vocabulary size: V
- Each word is mapped to an index:

$$w_{d,n} \in \{1, \dots, V\}$$

- Word counts:

$$x_{d,v} = \text{count of term } v \text{ in document } d$$

Example

Documents:

1. stephen is nice
2. john is also nice
3. george is mean

Vocabulary:

$$\{\text{stephen, is, nice, john, also, george, mean}\}$$

Index:

$$\text{stephen} = 1,\ \text{is} = 2,\ \text{nice} = 3,\ \text{john} = 4,\ \text{also} = 5,\ \text{george} = 6,\ \text{mean} = 7$$

Representation:

$$w_1 = (1, 2, 3), \quad w_2 = (4, 2, 5, 3), \quad w_3 = (6, 2, 7)$$

Document-Term Matrix

Idea

Represent text as a matrix of word counts

$$X = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

- Rows = documents
- Columns = words
- Entries = counts

From Document-Term Matrix to Models

- The **document-term matrix** is the foundation of much of text analysis in economics
- It represents text as counts of words across documents
- However, the **bag-of-words** approach ignores the dependence structure among words

Challenge

Text data is high-dimensional: thousands of words, sparse representation, and complex relationships.

- Goal: reduce dimensionality while preserving meaningful variation across documents
- Approach: **topic models** (factor models for discrete data)

Two Core NLP Problems

The problem of *synonomy* is that several different words can be share similar meanings. Cosine similarity between following documents?

school	university	college	teacher	professor
0	5	5	0	2

school	university	college	teacher	professor
10	0	0	4	0

The problem of *polysemy* is that the same word can have multiple meanings. Cosine similarity between following documents?

tank	seal	frog	animal	navy	war
10	10	3	2	0	0

tank	seal	frog	animal	navy	war
10	10	0	0	4	3

Where Economists Use Text Data

Source	What economists use it for
FOMC statements	Measuring monetary policy *tone* and hawkishness
SEC 10-K / 10-Q filings	Detecting firm-level risk and uncertainty
Earnings call transcripts	Management sentiment, forward guidance
News articles	Event studies, media bias, policy salience
Job postings	Labor market demand, skill requirement shifts
Congressional records	Policy attention, partisan divergence
Social media	Consumer sentiment, crisis early warning

Key insight

Word choice is **behavior**. When the Fed says "restrictive" instead of "accommodative," that is a policy signal, measurable, repeatable, and analyzable.

Opening Puzzle: Two Sentences, One Meeting

Dovish (2020)

"The Committee will maintain the current accommodative stance of monetary policy to promote maximum employment and stable prices."

(2022)

"The Committee is strongly committed to returning inflation to its 2 percent objective and will act forcefully to restore price stability."

Both are one sentence. Both discuss Fed policy.

How do we teach a machine to understand the difference?

That is the question this lecture answers.

The Text-to-Numbers Pipeline

Raw text → **1. Clean** → **2. Tokenize** → **3. Remove stops** → **4. Vectorize**

Clean
Lowercase, strip punctuation & digits

Tokenize
Split into individual words

Remove stopwords
Drop "the", "a", "is", "of"

Vectorize
Word counts, TF-IDF, embeddings

Example: one sentence through the pipeline

Raw: "The Committee will maintain the current accommodative stance."

Clean: "the committee will maintain the current accommodative stance"

Tokenize: ["the", "committee", "will", "maintain", ..., "stance"]

Remove stops: ["committee", "maintain", "current", "accommodative", "stance"]

Why build it step by step?

In production, `TfidfVectorizer` does all four steps in one call. But black-box tools produce black-box errors, understanding each step separates practitioners from button-pushers.

Step 1. Cleaning and Tokenization

The cleaning function

```
def clean_text(text):
    text = text.lower()
    text = re.sub(r'[^a-z\s]', ' ',
        text)
    text = re.sub(r'\s+', ' ', text).
        strip()
    return text
```

What gets removed

- 1/4, 2, 10-K, digits and special chars
- `Inflation` → `inflation`, case folding
- Extra whitespace, newlines

Tokenization

```
from nltk.tokenize import
    word_tokenize
tokens = word_tokenize(
    clean_text)
```

Before:

"The Fed raised rates to 1/4%."

After tokenize + stopword removal:

`['fed', 'raised', 'rates']`

Rule of thumb

≈ 40–50% of tokens are stopwords.
What survives carries the signal.

Step 2. Word Frequency (Bag of Words)

The idea: throw all tokens into a bag and count.

Top words across our FOMC corpus

Word	Count	Signal
inflation	42	hawkish
committee	38	neutral
employment	28	dovish
price	26	hawkish
accommodative	18	dovish
restrictive	12	hawkish
transitory	8	dovish

What frequency tells you

Which topics are discussed,
not how the author feels about them.

The critical limitation

"Inflation" appearing 50 times just means
inflation is the topic, it does **not** tell you if the Fed is calm or alarmed.

We need **sentiment** for that.

Still useful for:

- Topic detection
- Tracking word use over time
- Finding era-specific terminology

TF-IDF: Weighting Words

Idea

Not all words are equally informative. TF-IDF gives more weight to words that are important in a document but rare across documents.

Two Components

- **Term Frequency (TF)** How often a word appears in a document
- **Inverse Document Frequency (IDF)** Downweights words that appear in many documents

$$\text{TF-IDF}_{d,v} = \text{TF}_{d,v} \times \log\left(\frac{D}{\text{DF}_v}\right)$$

Intuition

Words like *the, is, and* get low weight. Words like *inflation, recession* get higher weight.

Step 3. TF-IDF: What Makes a Document Distinctive?

The core question

Which words are distinctive to *this* document compared to all others?

The formula:

$$\text{TF-IDF}(w,d) = \underbrace{\frac{\text{count}(w,d)}{\text{words in } d}}_{\text{TF}} \times \underbrace{\log\left(\frac{N}{\text{docs with } w}\right)}_{\text{IDF}}$$

TF How often does word w appear *in this document*?

IDF How rare is w *across all N documents*?
Rare $\Rightarrow$ higher weight.

High TF-IDF score

Appears **often in this doc** but **rarely elsewhere**

Low TF-IDF score

Appears in every document (e.g., "committee", "policy")

Key advantage

No manual stopword list needed, common words are down-weighted **automatically** by IDF.

TF-IDF in Action. Top Distinctive Words by Statement

Statement	Top distinctive words (TF-IDF)
2020-03 (Dovish)	**tools**, **challenging**, **accommodative**, **purchase**, **range**
2021-06 (Dovish)	**transitory**, **vaccinations**, **bottlenecks**, **patient**, **normalization**
2021-09 (Taper)	*moderation*, *warranted*, *pace*, *imbalances*, *broadly*
2022-03 (Hike)	**attentive**, **tighten**, **balance sheet**, **sheet**, **firming**
2022-09 (Restrict)	**restrictive**, **sufficiently**, **lags**, **cumulative**, **attain**
2022-12 (Peak)	**resolved**, **forcefully**, **persist**, **tightening**, **essential**
2023-06 (Pause)	*steady*, *assess*, *implications*, *monitor*, *firming*

The key insight

TF-IDF does what a human analyst would do:
find the words that make each document *unique*.

Limitation

TF-IDF still does not know
what is being said about those words,
only that they are *distinctive*.

Step 4. Sentiment Analysis with VADER

What VADER is

- Rule-based lexicon: $\sim$7,500 hand-labeled words
- Returns 4 scores per document:
  - `pos`: fraction of positive text
  - `neg`: fraction of negative text
  - `neu`: fraction of neutral text
  - `compound`: overall score $\in [-1, +1]$

```
from nltk.sentiment.vader
    import (
     SentimentIntensityAnalyzer
)
sia =
    SentimentIntensityAnalyzer
    ()
score = sia.polarity_scores(
    text)
% Returns: {pos:0.18, neg:0.04,
%                neu:0.78, compound
    :0.42}
```

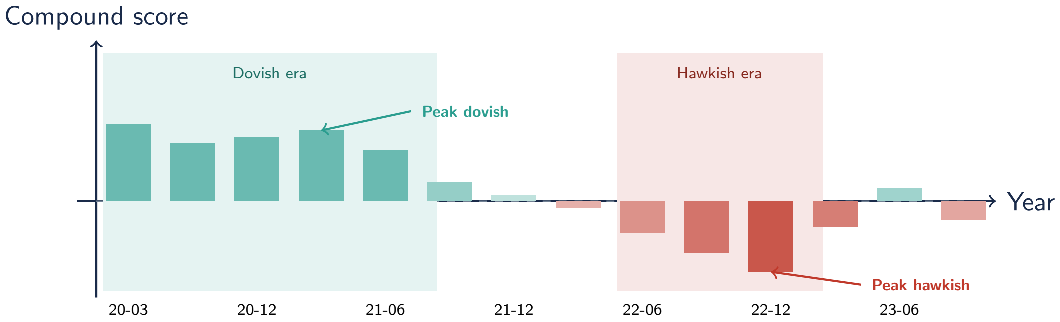
Compound score guide

$> +0.05$	Positive / optimistic
< -0.05	Negative / pessimistic
In between	Neutral

Important

Run VADER on **raw text**, not cleaned. VADER uses capitalization and

The FOMC Sentiment Arc, 2020–2023



Pattern: Sentiment is positive during the dovish era (2020–21), declines sharply through 2022, troughs at "Peak Hawkishness" (2022-12), then partially recovers as language shifts toward pausing in 2023.

Break the Analyzer. When VADER Gets the Fed Wrong

VADER was trained on general English. It does **not** know Fed-speak.

Phrase	VADER	Economic meaning
"sufficiently restrictive stance"	**neutral/positive**	Hawkish – signals hikes
"strongly committed to price stability"	**positive**	Hawkish – inflation priority
"act forcefully to restore stability"	**positive!**	Hawkish – aggressive hikes
"solid and robust pace of expansion"	**positive**	No cuts – policy unchanged
"patient and accommodative approach"	**positive**	Dovish – correct signal

Root cause

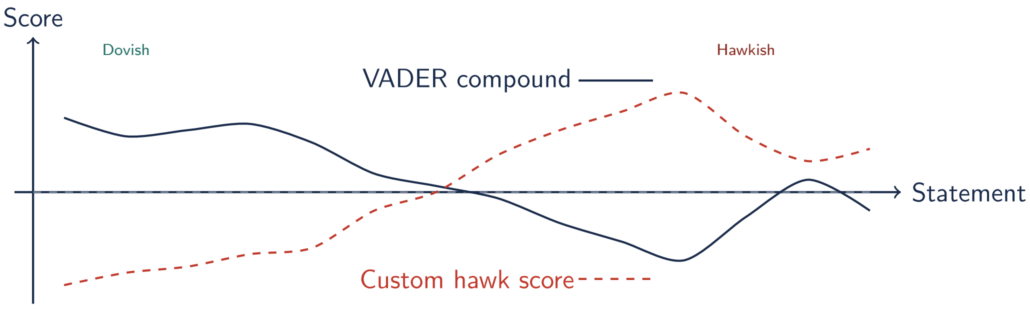
"Restrictive", "committed", "forcefully", "stability", all sound fine in everyday English. VADER does not know they are **hawkish signals**.

The fix: domain adaptation

Build a custom word list of hawk/dove signal words.

hawk score = (hawk count – dove count) / total words

VADER vs. Domain-Adapted Score



Key result

The custom domain-adapted score shows a **cleaner dovish-to-hawkish arc** than VADER, which is distorted by general-purpose vocabulary mismatches.

From Words to Regression Inputs

The bridge to what you already know

NLP outputs are numbers. Numbers go into regressions.

NLP output as outcome variable:

- Does Fed sentiment predict yield changes?
- Do 10-K risk words predict stock returns?
- Does earnings-call tone predict next-quarter revenue?

```
model = smf.ols(
  'yield_change ~ vader_compound',
  data=df_fomc_merged
).fit()
```

NLP output as explanatory variable:

- Does uncertainty language (X) slow hiring (Y)?
- Does CEO tone predict firm investment?

Causal inference warning

Significance $\neq$ causation.
Text scores are observational.
You still need an identification strategy.

The full pipeline we built today

Key Takeaways

1. **Text is data you haven't converted yet.** Every NLP step makes one piece of that conversion explicit.
2. **Word frequency tells you topic, not tone.** "Inflation" appearing 50 times $\neq$ the Fed is alarmed.
3. **TF-IDF rewards distinctiveness, not frequency.** Common words are penalized automatically.
4. **VADER is general-purpose, domain adaptation matters.** "Forcefully" sounds positive in everyday English. In a Fed statement, it means aggressive rate hikes.
5. **NLP outputs are regression inputs.** Sentiment scores and TF-IDF vectors are numbers. The bridge from text to causal inference is shorter than it looks.